PRANOVA

Healthcare Coordination Platform & Clinical Operations Engine

Document	Client Operations & Administration User Guide
Version	2.0 (Production Release)
Date	October 2026
Audience	Platform Administrators, Operations Supervisors, Clinical Coordinators
Architecture	Strictly Headless API-First, MySQL 8 / Prisma, React-Admin v5
Deployment	Production Single-Container (Railway, Docker, Cloud VMs)
Classification	CONFIDENTIAL — Prepared for client operations. Not for public distribution.

Executive Overview: PRANOVA provides an enterprise healthcare operating system connecting Central Administrators, Patients, Field Care Managers, and Healthcare Providers. This document provides comprehensive operational guidance for managing clinician rosters, appointments, online pharmacy orders, diagnostic pathology tests, and emergency SOS cases.

1. Welcome to PRANOVA

PRANOVA is an enterprise-grade, headless, API-first healthcare coordination platform. A single robust Node.js Express backend engine connects three distinct user surfaces:

- **Central Admin Web Console:** A high-density clinical administration portal built with React-Admin v5 and Material-UI v5 for operators, supervisors, and clinical coordinators.
- **Patient Mobile App:** A native React Native (Expo) consumer application for doctor discovery, appointments, medicine orders, lab tests, and emergency SOS dispatching.
- **Care Manager Mobile App:** A native React Native (Expo) field application for clinical officers managing encounters, accepting emergency triage, and updating case timelines.

2. Production Architecture & Technology Stack

The platform is locked onto standard, enterprise technologies designed for high reliability, strict referential integrity, and sub-millisecond query execution:

Layer	Technology	Production Specification
Database	MySQL 8.0+ / MariaDB	Port 3306 (pranova_db) with Prisma ORM & strict foreign keys
Backend API	Node.js / Express / TypeScript	RESTful JSON APIs (/api/v1/*) with Zod request validation
Real-Time	Socket.IO Engine	Bidirectional emergency telemetry & audio dispatch alerts
Admin Console	React-Admin v5 (Vite + MUI)	Served directly by Express in production from / and /admin
Mobile Apps	React Native (Expo SDK 57)	Native mobile apps consuming dedicated /app/* and /cm/* APIs

Unified Single-Container Serving

In production, the backend server automatically detects and serves the compiled React-Admin bundle at / and /admin. This enables single-origin deployment without CORS issues or the need for separate frontend hosting servers. The same container provides the REST API, WebSockets, and the operational Web Console.

3. Accessing the Admin Console

Administrators access the central console through any standard modern web browser:

- **Local / Dedicated Server:** http://localhost:5000/ or /admin
- **Railway / Cloud Hosting:** Your generated cloud domain (e.g. https://pranova-production.up.railway.app)

Parameter	Default Value	Configuration Note
Super Admin Email	admin@pranova.com	Configurable via ADMIN_EMAIL in .env
Default Password	Admin@123456	Configurable via ADMIN_PASSWORD in .env
Landing View	#/dashboard	Command Center Operations Dashboard

4. The Command Center Dashboard

The **Command Center** (#/dashboard) delivers real-time operational telemetry across the entire healthcare ecosystem. Each metric tile functions as an interactive gateway to its resource:

Metric Tile	Clinical / Operational Meaning	Target Action
Revenue (GMV)	Total combined revenue across consultations, pharmacy, and lab tests.	Monitors platform commercial health
Active Doctors	Total clinicians currently accepting patient appointments.	Onboard or update clinician rosters
Appointments	Total booked visits, highlighting today's scheduled consultations.	Review pending appointment confirmations
Pharmacy Orders	Total prescription orders with active fulfillment count.	Process packing and dispatching
Lab Bookings	Diagnostic tests awaiting sample collection or report release.	Upload verified PDF laboratory reports
Registered Patients	Total verified patient profiles in the clinical directory.	Inspect clinical history & allergies
Active Cases	Emergency SOS and clinical cases requiring care coordinator action.	Immediate field care manager triage

5. Managing Specialties & Doctors

5.1 Specialties Catalog (#/specialties): Configure medical specialties browsed by patients in the mobile app (e.g. Cardiology, Pediatrics, General Medicine, Dermatology, Orthopedics). Each specialty carries an icon, title, and description.

5.2 Doctor Directory (#/doctors): Manage clinician credentials and availability:
• **Profile:** Full name, medical qualifications (MBBS, MD), registration license, years of experience, languages, and biography.
• **Consultation Fees:** Separate pricing for in-clinic visits and virtual teleconsultations.
• **Availability Slots:** Configure weekly recurring consultation hours, consultation duration, and room type.
• **Status:** Switch clinician between ACTIVE, ON_LEAVE, and INACTIVE.

6. Appointments & Video Teleconsultations

The appointment engine coordinates in-clinic and video consultations through a verified lifecycle:

Status	Workflow Stage & Operational Meaning	Actor Responsible
REQUESTED	Patient booked a consultation slot via the mobile app.	Patient / System
CONFIRMED	Slot approved; encrypted video room automatically generated for teleconsults.	System / Admin
IN_PROGRESS	Clinician and patient are actively conducting consultation.	Doctor / Coordinator
COMPLETED	Consultation concluded; prescription and clinical notes uploaded.	Doctor / Coordinator
CANCELLED	Appointment cancelled before start time.	Patient / Admin
NO_SHOW	Patient failed to attend scheduled in-clinic or video session.	Doctor / Coordinator

Secure Video Rooms: For teleconsultations, PRANOVA issues an encrypted video room link. Both patient and clinician enter with signed tokens, ensuring privacy compliance under DPDP and HIPAA.

7. Online Pharmacy & Medicine Logistics

7.1 Medicine Catalog (#/medicines): Add brand and generic pharmaceuticals with dosage forms (Tablets, Syrups, Injections, Ointments), strengths, retail MRP, and prescription requirements.

7.2 Order Fulfillment Pipeline (#/orders): Orders progress through a 5-stage dispatch pipeline:

Order Status	Fulfillment Action Required	Automated Operations
PLACED	Order submitted by patient; prescription attached if required.	Inventory reserved; server calculates totals.
CONFIRMED	Pharmacist verifies prescription and accepts order.	Payment captured / COD authorized.
PACKED	Medications boxed with invoice and batch verification.	Dispatch label generated.
OUT_FOR_DELIVERY	Order assigned to courier fleet for doorstep delivery.	Patient receives dispatch notification.
DELIVERED	Courier confirms delivery at patient physical address.	Order marked completed and paid.
CANCELLED	Order rejected or cancelled by administrator.	Reserved inventory returned to catalog.

8. Diagnostics & Laboratory Bookings

8.1 Test Catalog (#/diagnostic-tests): Pathology panels (CBC, Lipid Profile, HbA1c, etc.) with fasting guidelines, sample types (Blood, Urine, Swab), and turnaround hours.

8.2 Lab Booking Workflow (#/lab-bookings): Patients choose Home Collection or Lab Visit:

Status	Investigation Stage	Clinical Responsibility
BOOKED	Patient booked diagnostic package via mobile app.	Phlebotomist scheduled for dispatch
SAMPLE_COLLECTED	Specimen collected from patient and barcoded.	Phlebotomist / Collection Center
PROCESSING	Laboratory analyzers conducting biochemical analysis.	Pathology Laboratory Technician
REPORT_READY	Certified clinical PDF report attached to booking.	Released to patient mobile app
CANCELLED	Investigation cancelled prior to specimen collection.	Patient / Administrator

9. Emergency SOS Cases & Field Triage

When a patient triggers the emergency **SOS Button** in their mobile app:

1. An immutable **P1 Emergency Case** is created with exact GPS coordinates and physical street address.
2. The backend broadcasts a real-time admin:new_emergency event over WebSockets.
3. The Admin Console sounds an audible chime and displays the **Emergency Dispatch Modal**.
4. The dispatcher reviews the triage map, acknowledges the incident, and assigns a field Care Manager.

Priority	Clinical Severity & Criteria	SLA & Protocol
P1 — Critical	Acute life threat (chest tightness, dyspnea, stroke symptoms, major trauma).	Immediate field triage dispatch (< 5 mins)
P2 — Urgent	Severe unstable symptoms (high fever, severe abdominal pain, post-op crisis).	Field clinical response (< 30 mins)
P3 — Standard	Routine care coordination, medication review, post-discharge check-in.	Scheduled care manager encounter

Interactive Location Link with OpenStreetMap

All incident locations, clinic addresses, and delivery addresses render through LocationLink. Clicking an address opens an interactive OpenStreetMap dialog showing the exact geographic pin, with one-click links to Google Maps and OpenStreetMap navigation directions.

10. Patient & Provider Registries

• **Patients (#/patients):** Complete patient registry with blood group, documented allergies, emergency contacts, linked family members, and clinical documents.
• **Care Managers (#/care-managers):** Field healthcare officers with assigned zones, active case counts, and contact hotlines.
• **Providers (#/providers):** Network of certified Hospitals, Clinics, Diagnostic Labs, Pharmacies, and Ambulance fleets with GPS mapping.

11. Clinical Document Management

The document subsystem (#/documents) securely manages prescriptions, lab test reports, and discharge summaries. Documents are accessed via signed token endpoints with automatic content-type detection and verified clinical PDF previews.

12. Platform Branding & Runtime Settings

Administrators customize the platform runtime via **Settings** (#/settings) without code modifications:

- **Application Name & Tagline:** Displayed on the browser title bar, login portal, and AppBar.
- **Primary Brand Color:** Controls AppBar styling, primary action buttons, and status chips.
- **Logo & Favicon URLs:** Custom image assets for your healthcare organization.
- **Support Contact:** Official support email and 24/7 emergency dispatch hotline numbers.

13. Production Cloud Deployment (Railway & Docker)

PRANOVA includes zero-config production deployment specifications for cloud container platforms:

Step	Railway / Cloud Action	Automated Platform Execution
1. Connect Repo	Connect GitHub repo to Railway project.	Nixpacks detects Node.js & railway.json.
2. Add MySQL	Add managed MySQL 8 database service.	Railway sets DATABASE_URL variable automatically.
3. Build	npm run build	Compiles backend TypeScript & React-Admin Vite UI.
4. Start	npm start	Syncs Prisma schema and boots server on port 5000.
5. Live Demo	Open generated public Railway domain.	Admin Console loads immediately at root URL.

Appendix A — First-Login Operational Checklist

Follow this operational verification checklist when launching a fresh PRANOVA deployment:

- **[✓] 1. Sign In:** Access the console with admin@pranova.com / Admin@123456.
- **[✓] 2. Update Security:** Navigate to Administrators and update the default password.
- **[✓] 3. Set Branding:** Configure organization name, brand color, and support email in Settings.
- **[✓] 4. Verify Specialties:** Confirm standard medical categories (Cardiology, Dermatology, etc.) exist.
- **[✓] 5. Onboard Doctors:** Add clinicians with qualifications, consultation fees, and weekly slots.
- **[✓] 6. Review Catalog:** Verify pharmaceuticals in the Medicine Catalog and laboratory tests in Diagnostics.
- **[✓] 7. Confirm Dashboard:** Verify the Command Center dashboard displays active metrics and charts.

Appendix B — Support & Engineering Assistance

For technical support, custom clinical workflows, or cloud infrastructure assistance:

Channel	Contact & Access Details
GitHub Repository	https://github.com/Aman-Kumar-roy/pranova
Support Email	support@pranova.health / engineering@pranova.com
API Documentation	Interactive OpenAPI Explorer at /api-docs (when enabled)
Licensing	MIT Open Source License

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